Getting started exercises

All questions are original. Record 3a means key 3 and identity a. Unless specified otherwise, use the chapter's 1-based indices, short-circuit semantics, and word-RAM model.

02-E01 Full insertion trace and separate counts

Run INSERTION-SORT on [4a,1b,3c,1d]. For each insertion list the saved key, all key comparisons, the hole representation after every shift, and the filled array. Count shifts, key comparisons, complete inner guards, and array writes. Check stability.

02-E02 Inner preservation and a counterexample

A proof claims that after each assignment A[i+1]=A[i], A[1..j] remains a permutation of the original prefix. Give a smallest counterexample. State the correct invariant using a hole and saved key, and prove initialization, maintenance, and exit. If > changes to $\geq$, which sorting properties survive and which fail?

02-E03 Shifts count inversions

Prove $\sum h_j = I$ for general input. Compute each h_j and I for [3a,1b,3c,2d]. Find E[I] under uniform permutations of n distinct records and explain why indicator independence is unnecessary. What is E[I] after uniformly permuting record identities when all keys are equal?

02-E04 A distribution for average search cost

Linear search reads n distinct keys in order and returns at a match. Success has probability s; conditional on success, positions are uniform. Failure reads all n keys. Derive expected key comparisons and evaluate n=8,s=3/4. How does s affect the worst-case bound?

02-E05 Complete stable MERGE

Merge L=[0a,3b,7c] and R=[1d,3e,6f]. List i,j before each comparison, output prefixes, and the exhaustion tail. Count comparisons, buffer copies, and output writes. If ties choose R instead, what output results, and is it a correct sorting?

Getting started exercises (continued)

02-E06 Sentinels and aliases have assumptions

Legal keys include maximum M. Trace a sentinel implementation using M on L=[M],R=[M,M], with exactly m outputs, through the out-of-bounds read. A second implementation uses a view of A's left half instead of a copy when merging A=[2,4,1,3]. Identify the first destructive overwrite and later bad read. Give a repair without changing the key domain.

02-E07 Odd lengths and peak space

For [6a,2b,5c,1d,4e], list every (p,q,r) and merged result. Count cumulative buffer-copy records, output writes, and maximum records buffered in one merge. Distinguish cumulative allocation, peak buffers, and the call stack.

02-E08 Exact recurrence solutions

For $n = 2^h$, solve $U(1) = 3, U(n) = 2U(n/2) + 5n$ and $W(1) = 0, W(n) = 2W(n/2) + n - 1$ exactly by expansion or induction. Evaluate at 8. W uses the worst comparison count at every merge. Why is it not the exact count for every input?

02-E09 Θ bounds beyond powers of two

Without the master theorem, prove Θ(n log n) for every $n \geq 2$. Specify interval sizes, upper and lower level costs, stopping depth, and total base-case costs. Rounding n up to a power of two and assuming time monotonicity is insufficient.

02-E10 Model changes and an early-exit variant

For sorted input, give insertion-sort and standard merge-sort orders. After recursively sorting the two halves, a variant skips MERGE if A[q]≤A[q+1]. Prove correctness and derive sorted-input time. If a key comparison requires Θ(L) character reads, which constant-cost conclusions need revision?

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study